

For exercises 1-2, let $f(x) = 7 - 5x$ and $g(x) = 2x - 3$. Find the following.

1. $f(2)$

2. $g(1.5)$

Solve the problem.

3. Let the supply and demand functions for butter pecan ice cream be given by

$$p = S(q) = \frac{2}{5}q \text{ and } p = D(q) = 100 - \frac{2}{5}q,$$

where p is the price in dollars and q is the number of 10-gallon tubs. Find the equilibrium quantity and the equilibrium price.

4. FedUps delivers packages which cost \$2.00 per package to deliver. The fixed cost to run the delivery truck is \$324 per day. If the company charges \$5.00 per package, how many packages must be delivered to make a profit of \$42?

SHORT ANSWER. Write the word or phrase that best completes each statement or answers the question.

5. In one or more paragraph, explain what it means to “break even” in business.

Find the equation of the least squares line.

6. The paired data below consists of the average pupil-teacher ratio in public schools for 5 randomly selected years.

Year	1990	1994	1998	2002	2006
Ratio	17.4	17.7	16.9	16.2	15.8

7. The paired data below consists of customers (in millions) for 5 selected years.

Year	2000	2002	2004	2006	2008
Customers	8.5	19.3	25.4	32.6	40.4

SHORT ANSWER. Write the word or phrase that best completes each statement or answers the question.

8. Which sets of paired data from problems 6 and 7 have a stronger linear relation? (i.e., of the two equations which gives a better prediction for future values?)